

Conclusion

CONCLUSION • THE ONE MESSAGE

Correlation → Causation

... and from observation to intervention — the shift from data-driven (DDSE) to causal (Causal Software Engineering) software engineering.

Present • DDSE (ML)

Pattern recognition — associations in data

Correlation — acts on (maybe spurious) patterns

Observation — learns $P(X | W)$

Opaque — limited explainability

Implicit — assumptions hidden in data

**Future • CSE (Causal)**

Causal inference — what-if, explanations, hypotheses

Causation — confounder-free decisions

Intervention — learns $P(X | \text{do}(W))$

Transparent — explainable reasoning paths

Explicit — human knowledge in the model

THE ROAD AHEAD · FOUR CO-EVOLVING ROUTES

Where causal software engineering is heading

**R1 · Causal observability**

Augment metrics, logs & traces with explicit causal structure

Dream: reliable why & what-if answers

**R2 · Intervenability by design**

Treat CI/CD, flags & canaries as explicit causal interventions

Dream: evidence-backed intervention planning

**R3 · Counterfactual assurance**

Counterfactual RCA at scale; causal testing; refutation

Dream: routine “what would have avoided this?”

**R4 · Governance & alignment**

Auditable causal claims; human-in-the-loop; constrained agents

Dream: governed causal copilots

Roadmap adapted from: R. Pietrantuono, L. Giamattei, S. Russo, J. Siebert, N. Walkinshaw — “Causal Software Engineering: A Vision and Roadmap”, FSE 2026 – IVR track 2026 (arXiv:2605.02454).

CONCLUSION · WHERE WE ARE → WHAT'S NEXT

A growing footprint — and the path to adoption

Causal methods are spreading across SE; turning them into everyday practice is the next step.

Where we are

- 90+ SE studies now map to the Causal SE framework
- Strongest in fault localization, testing & analysis, prediction
- Tooling maturing: PyWhy (DoWhy, EconML), DagTTY, Tetrad
- Mostly reliability & performance — security & safety still rare

What's next

- Scaffold causal thinking — tutorials, examples, case studies
- Make refutation & robustness tests standard practice
- Integrate as an augmentation layer over CI/CD and ML tooling
- Pair causal reasoning with LLMs & agents — human in the loop

Source: CACM/ACM 2025 “Where we are, what’s next” — Pietrantuono, Giamattei & Russo; mapping studies Giamattei et al. 2025.

CONCLUSION · KEEP GOING

Continue your causal journey

Pointers to go deeper — books, lectures, and tools to start (or continue) your causal journey.

Books

- Pearl & Mackenzie — The Book of Why
- Hernán & Robins — Causal Inference: What If
- Huntington-Klein — The Effect (free book)

Lectures & courses

- Brady Neal — Introduction to Causal Inference (free)
- B. Neal & J. Pearl — video lecture series
- “A Crash Course in Causality” (Coursera, Penn)

Tools & community

- PyWhy: DoWhy, EconML, causal-learn
- Tetrad (CMU) — causal discovery
- DAGitty — draw & test DAGs